

Deep Generative Models for Density Estimation and Unpaired Image Translation: A Comprehensive Study of MAF and CycleGAN

*Note: Assignment 2 for Deep Generative Models Course

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Abstract—This paper investigates two complementary deep generative paradigms: (i) Masked Autoregressive Flow (MAF), a likelihood-based normalizing flow enabling explicit density estimation and anomaly scoring, and (ii) CycleGAN, an adversarial framework for unpaired image-to-image translation leveraging cycle consistency and identity constraints. We present unified theoretical derivations including change of variables formulations, Jacobian determinant computations, and autoregressive factorizations. Architectural design rationales are detailed for MADE masking schemes, ResNet-based generators, and PatchGAN discriminators. Training methodologies encompass optimization objectives and replay buffer stabilization techniques. Empirical validation is conducted through density modeling on industrial capsule imagery (MVTec AD dataset) and stylistic horse-zebra translation, evaluated via AUROC metrics (0.72 achieved) for anomaly detection and qualitative assessments for translation fidelity. Critical analysis addresses fundamental limitations including likelihood calibration challenges and geometric transformation constraints. Ethical deployment considerations and future research directions are systematically examined. This work consolidates foundational theory, implementation specifics, experimental results, and actionable insights for practitioners seeking robust generative modeling solutions.

Index Terms—Deep generative models, normalizing flows, masked autoregressive flow, CycleGAN, unpaired image translation, anomaly detection, cycle consistency, density estimation

I. INTRODUCTION

DEEP generative models have revolutionized the ability to learn complex data distributions and perform cross-domain transformations. Flow-based models, exemplified by Masked Autoregressive Flows (MAF), provide tractable likelihood computation essential for statistical inference tasks including anomaly detection and density estimation. Adversarial translation frameworks, particularly CycleGAN, emphasize perceptual realism in conditional generation without requiring paired supervision. This paper examines MAF and CycleGAN as representative instances of likelihood-based and adversarial paradigms respectively, providing technically rigorous yet implementation-focused analysis suitable for reproducible experimentation.

A. Motivation and Context

Industrial defect detection systems benefit from explicit probability density estimation, enabling prioritization of data regions exhibiting atypical statistical signatures. Manufacturing quality control demands automated anomaly identification with quantifiable uncertainty. In artistic, biological, and medical imaging domains, scarcity or infeasibility of paired training examples necessitates unpaired translation frameworks. Cross-modal synthesis (e.g., MRI-CT translation, seasonal style transfer) requires learning mappings between domains without pixel-level correspondence. Understanding both likelihood-based and adversarial approaches enables hybrid architectures combining their complementary strengths—explicit probabilistic scoring with perceptually realistic generation.

B. Paper Contributions

This work provides comprehensive treatment of two foundational generative modeling paradigms through:

- Unified mathematical derivations for coupling and autoregressive transformations with computational complexity analysis and practical implementation implications
- Detailed architectural specifications including layer-by-layer parameter counts, activation functions, normalization schemes, and design trade-offs
- Structured evaluation methodology employing threshold-independent metrics (AUROC) for anomaly detection and qualitative/quantitative assessments for image translation
- Systematic dissection of CycleGAN training objectives, stability mechanisms (replay buffers, learning rate scheduling), and inherent constraints on geometric transformations
- Critical examination of failure modes, likelihood calibration challenges, and ethical deployment considerations
- Reproducibility documentation with complete hyperparameter specifications, preprocessing pipelines, and computational resource requirements

II. BACKGROUND

A. Normalizing Flows: Mathematical Foundation

Normalizing flows learn an invertible mapping $f_\theta : X \rightarrow Z$ pushing data x to latent z under a simple base density p_Z (typically isotropic Gaussian). The log-likelihood decomposes across successive bijections:

$$\log p_\theta(x) = \log p_Z(f_\theta(x)) + \log \left| \det \frac{\partial f_\theta(x)}{\partial x} \right|. \quad (1)$$

a) *Composing Flows*: Given sequence $f = f_K \circ \dots \circ f_1$, Jacobians multiply yielding:

$$\log p_\theta(x) = \log p_Z(z_K) + \sum_{k=1}^K \log \left| \det \frac{\partial f_k}{\partial z_{k-1}} \right|, \quad z_k = f_k(z_{k-1}). \quad (2)$$

Each bijection f_k must have tractable Jacobian determinant computation (ideally $\mathcal{O}(D)$ rather than $\mathcal{O}(D^3)$ for full matrix determinant).

b) *Invertibility Requirements*: Training via maximum likelihood requires forward pass (data $\rightarrow$ latent) for density evaluation; sampling requires inverse (latent $\rightarrow$ data). Coupling layers provide both directions efficiently; autoregressive flows favor one direction.

B. Autoregressive Modeling and MADE

Autoregressive factorization decomposes joint density:

$$p(x) = \prod_{d=1}^D p(x_d \mid x_{<d}), \quad (3)$$

where each conditional $p(x_d \mid x_{1:d-1})$ may be parameterized by neural networks. MADE (Masked Autoencoder for Distribution Estimation) enforces this dependency structure via binary masks M applied to weight matrices, zeroing connections from x_j to x_i whenever $j \geq i$ under the chosen ordering.

a) *Masking Strategy*: Assign each hidden unit h a random number $m(h) \in \{1, \dots, D-1\}$ representing maximum input dependency. Mask from input x_d to hidden h is 1 iff $d \leq m(h)$; mask from hidden h to output parameter for x_d is 1 iff $m(h) < d$. This guarantees autoregressive property.

b) *Connection to MAF*: MAF stacks MADE blocks where each outputs transformation parameters (μ_d, σ_d) depending autoregressively on previous dimensions. Permuting dimension order between blocks increases expressiveness by varying dependency patterns.

C. Adversarial Image Translation

Generative adversarial networks pit generator G_θ against discriminator D_ϕ in a minimax game:

$$\min_{\theta} \max_{\phi} \mathbb{E}_{x \sim p_{\text{data}}} [\log D_\phi(x)] + \mathbb{E}_{z \sim p_Z} [\log(1 - D_\phi(G_\theta(z)))]. \quad (4)$$

a) *Conditional GANs (cGANs)*: Conditioned on input x , generator produces $G(x)$ targeting output domain; discriminator evaluates pairs (x, y) . Pix2pix exemplifies supervised paired translation with $\mathcal{L}_1$ reconstruction penalty.

b) *Unpaired Translation Challenge*: CycleGAN addresses absence of paired (x, y) by learning dual mappings ($G : X \rightarrow Y, F : Y \rightarrow X$) with cycle consistency constraint $F(G(x)) \approx x$ and $G(F(y)) \approx y$, preventing mode collapse while avoiding explicit correspondence.

D. Instance Normalization vs Batch Normalization

Instance normalization (IN) normalizes each sample independently across spatial dimensions:

$$\text{IN}(x) = \gamma \frac{x - \mu(x)}{\sigma(x)} + \beta, \quad (5)$$

where μ, σ computed per-instance per-channel. This contrasts with batch normalization (BN) using mini-batch statistics. IN better preserves individual image style statistics, critical for style transfer tasks.

E. Residual Learning

Residual blocks learn additive corrections $\mathcal{F}(x)$ to input:

$$y = x + \mathcal{F}(x), \quad (6)$$

mitigating vanishing gradients in deep networks. For image translation, residuals encode localized texture modifications while identity path maintains structure.

III. METHODOLOGY: DETAILED IMPLEMENTATION

A. MAF Architecture Specification

1) *MADE Block Structure*: Each MADE block consists of:

- **Input layer**: $D = 49,152$ dimensions (flattened $128 \times 128 \times 3$ RGB images).
- **Hidden layers**: Two fully-connected layers with 512 units each, masked weights enforcing autoregressive property.
- **Output layer**: $2D$ outputs encoding per-dimension mean $\mu_d(x_{<d})$ and log-scale $\log \sigma_d(x_{<d})$.
- **Activation**: ReLU for hidden layers; linear output.

2) *Transformation Equation*: For dimension d in block k :

$$z_d^{(k)} = \frac{x_d^{(k)} - \mu_d^{(k)}(x_{<d}^{(k)})}{\sigma_d^{(k)}(x_{<d}^{(k)})}, \quad (7)$$

where $\mu_d^{(k)}, \sigma_d^{(k)}$ are neural network outputs. Inverse for sampling:

$$x_d^{(k)} = \mu_d^{(k)}(x_{<d}^{(k)}) + \sigma_d^{(k)}(x_{<d}^{(k)}) \cdot z_d^{(k)}. \quad (8)$$

3) *Multi-Block Composition*: Seven MAF blocks stacked with dimension permutations (reversal or random shuffle) between blocks. Final latent $z^{(7)}$ compared against standard Gaussian base. Total trainable parameters: approximately 18M (dominated by $512 \times 49,152$ weight matrices).

4) *Training Objective and Gradients*: Minimize negative log-likelihood:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \left[\log p_Z(z_i^{(K)}) + \sum_{k=1}^K \sum_{d=1}^D \log \sigma_d^{(k)} \right], \quad (9)$$

where $p_Z = \mathcal{N}(0, I)$. Gradients backpropagate through MADE networks updating $\mu, \log \sigma$ parameters.

B. CycleGAN Architectural Components

1) Generator Architecture: ResNet-9:

a) Downsampling Path::

- 1) Initial convolution: 7×7 kernel, 64 filters, Instance Norm, ReLU.
- 2) Downsample 1: 3×3 conv, stride 2, 128 filters, IN, ReLU.
- 3) Downsample 2: 3×3 conv, stride 2, 256 filters, IN, ReLU.

b) Transformation Path:: Nine residual blocks at resolution 64×64 , each comprising:

- 3×3 conv, 256 filters, IN, ReLU
- 3×3 conv, 256 filters, IN
- Residual connection (sum with input)

c) Upsampling Path::

- 1) Upsample 1: 3×3 transposed conv, stride 2, 128 filters, IN, ReLU.
- 2) Upsample 2: 3×3 transposed conv, stride 2, 64 filters, IN, ReLU.
- 3) Output convolution: 7×7 kernel, 3 filters (RGB), tanh activation $\rightarrow [-1, 1]$.

Total generator parameters: $\approx 11.4\text{M}$ per generator (G_{AB}, G_{BA}).

2) Discriminator Architecture: PatchGAN-70: Five convolutional layers with progressively increasing channels:

- 1) Conv: 4×4 , stride 2, 64 filters, LeakyReLU(0.2) — no normalization.
- 2) Conv: 4×4 , stride 2, 128 filters, IN, LeakyReLU.
- 3) Conv: 4×4 , stride 2, 256 filters, IN, LeakyReLU.
- 4) Conv: 4×4 , stride 1, 512 filters, IN, LeakyReLU.
- 5) Conv: 4×4 , stride 1, 1 filter (logit output) — no activation.

Output: 30×30 grid for 256×256 input (effective receptive field 70×70 per patch).

Total discriminator parameters: $\approx 2.8\text{M}$ per discriminator (D_A, D_B).

3) Loss Functions: Complete Formulation:

a) Adversarial Loss (LSGAN)::

$$\mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) = \mathbb{E}_{y \sim p_Y} [(D_Y(y) - 1)^2] + \mathbb{E}_{x \sim p_X} [D_Y(G(x))^2], \quad (10)$$

$$\mathcal{L}_{\text{GAN}}(F, D_X, Y, X) = \mathbb{E}_{x \sim p_X} [(D_X(x) - 1)^2] + \mathbb{E}_{y \sim p_Y} [D_X(F(y))^2]. \quad (11)$$

b) Cycle Consistency Loss::

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_X} [\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_Y} [\|G(F(y)) - y\|_1]. \quad (12)$$

c) Identity Loss::

$$\mathcal{L}_{\text{id}}(G, F) = \mathbb{E}_{y \sim p_Y} [\|G(y) - y\|_1] + \mathbb{E}_{x \sim p_X} [\|F(x) - x\|_1]. \quad (13)$$

d) Full Generator Objective::

$$\mathcal{L}_G = \mathcal{L}_{\text{GAN}}(G, D_Y) + \mathcal{L}_{\text{GAN}}(F, D_X) + \lambda_{\text{cyc}} \mathcal{L}_{\text{cyc}} + \lambda_{\text{id}} \mathcal{L}_{\text{id}}. \quad (14)$$

e) Discriminator Objectives::

$$\mathcal{L}_{D_Y} = \mathbb{E}_y [(D_Y(y) - 1)^2] + \mathbb{E}_x [D_Y(G(x))^2], \quad (15)$$

$$\mathcal{L}_{D_X} = \mathbb{E}_x [(D_X(x) - 1)^2] + \mathbb{E}_y [D_X(F(y))^2]. \quad (16)$$

C. Optimization Protocol

1) MAF Training:

- **Optimizer:** Adam with $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$.
- **Learning rate:** 10^{-4} (constant).
- **Batch size:** 3 (GPU memory limitation with 128^2 images).
- **Epochs:** 100.
- **Gradient clipping:** Norm threshold 5.0 to prevent NaN.
- **Early stopping:** Monitor validation NLL; halt if no improvement for 10 epochs.

2) CycleGAN Training:

- **Optimizer:** Adam with $\beta_1 = 0.5, \beta_2 = 0.999$ (lower momentum for adversarial stability).
- **Learning rate schedule:**
 - Epochs 1–100: Constant 2×10^{-4} .
 - Epochs 101–200: Linear decay to 0.
- **Batch size:** 1 (standard for unpaired translation to preserve individual statistics).
- **Update rule:** Alternate — update D_A, D_B then G_{AB}, G_{BA} per iteration.
- **Replay buffer:** Size 50; sample with 50% probability from buffer vs current fake.
- **Label smoothing:** Real labels sampled $\sim \mathcal{U}(0.9, 1.0)$; fake labels 0.

D. Implementation Details

a) Frameworks:: PyTorch 2.0+, torchvision for data loading.

b) Hardware:: Training conducted on NVIDIA GPU (CUDA-enabled); mixed precision (FP16) optional for memory efficiency.

c) Data Augmentation:: CycleGAN applies random horizontal flip; MAF uses raw images to preserve likelihood semantics.

d) Checkpointing:: Save model weights every 10 epochs; retain best based on validation metric (NLL for MAF, visual inspection for CycleGAN).

IV. HYPERPARAMETERS

V. EXPERIMENTAL SETUP AND PROTOCOLS

A. Datasets

1) *MVTec Anomaly Detection (Capsule Class)*: Industrial defect detection dataset containing:

- **Training:** 219 normal capsule images (defect-free).
- **Test:** 54 normal + 109 anomalous (various defect types: crack, poke, scratch, faulty imprint, squeeze).
- **Resolution:** Original high-resolution images.
- **Preprocessing:** Resize to 128×128 , normalize to $[0, 1]$, standardize per-channel (ImageNet statistics).

TABLE I: Key Hyperparameters for MAF and CycleGAN Models

Model	Parameter	Value	Rationale
MAF	Blocks	7	Expressiveness vs memory
MAF	Hidden width	512	Capacity for high D
MAF	Learning rate	1×10^{-4}	Stable NLL descent
MAF	Batch size	3	GPU memory constraints
CycleGAN	ResNet blocks	9	Recommended for 256^2
CycleGAN	Replay buffer size	50	Empirical stability
CycleGAN	λ_{cyc}	10	Balance realism/structure
CycleGAN	λ_{id}	5	Preserve coloration
CycleGAN	Optimizer	Adam	Widely adopted baseline

2) *Horse2Zebra*: Unpaired image translation dataset:

- **Domain A (Horse)**: 1,067 training images of horses in natural settings.
- **Domain B (Zebra)**: 1,334 training images of zebras.
- **Test sets**: 120 horses, 140 zebras.
- **Resolution**: Variable; resized to 256×256 with aspect-ratio-preserving crop.
- **Normalization**: Scale to $[-1, 1]$ to match tanh generator output range.

B. Data Preprocessing Pipeline

1) MAF Preprocessing:

- 1) Resize to 128×128 using bilinear interpolation.
- 2) Convert to tensor, scale pixel values $\in [0, 1]$.
- 3) Apply per-channel standardization: $x' = (x - \mu_{\text{ImageNet}}) / \sigma_{\text{ImageNet}}$.
- 4) Flatten to vector $\in \mathbb{R}^{49152}$ for flow input.
- 5) Add small noise ($\mathcal{N}(0, 10^{-3})$) for numerical stability (dequantization).

2) CycleGAN Preprocessing:

- 1) Load RGB image, resize shortest edge to 286.
- 2) Random crop to 256×256 (training); center crop (testing).
- 3) Random horizontal flip with probability 0.5 (training only).
- 4) Normalize to $[-1, 1]$: $x' = 2 \cdot (x/255) - 1$.

C. Training Procedures

1) MAF Training Algorithm:

```

for epoch = 1 to 100 do
  for each mini-batch  $\{x_i\}$  in training set do
    Forward pass:  $z = f_\theta(x)$ 
    Compute NLL:  $-\log p_\theta(x) = -\log p_Z(z) - \sum_d \log |\sigma_d|$ 
    Compute gradients:  $\nabla_\theta \mathcal{L}$ 
    Update parameters:  $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ 
  end for
  Evaluate validation NLL; checkpoint if improved
end for

```

2) CycleGAN Training Algorithm:

```

for epoch = 1 to 200 do
  for each pair  $(x, y)$  from unpaired dataloaders do

```

```

    Generate translations:  $\hat{y} = G_{AB}(x), \hat{x} = G_{BA}(y)$ 
    Compute cycles:  $\tilde{x} = G_{BA}(\hat{y}), \tilde{y} = G_{AB}(\hat{x})$ 
    Compute identities:  $\hat{y}_{\text{id}} = G_{AB}(y), \hat{x}_{\text{id}} = G_{BA}(x)$ 
    Query replay buffer:  $\hat{y}_{\text{old}} \sim \text{Buffer}_B(\hat{y})$ 
    Update  $D_A$ :  $\min \mathcal{L}_{D_A}(D_A, x, \hat{x}_{\text{old}})$ 
    Update  $D_B$ :  $\min \mathcal{L}_{D_B}(D_B, y, \hat{y}_{\text{old}})$ 
    Update generators:  $\min \mathcal{L}_G = \mathcal{L}_{\text{adv}} + \lambda_{\text{cyc}} \mathcal{L}_{\text{cyc}} + \lambda_{\text{id}} \mathcal{L}_{\text{id}}$ 
  end for
if epoch > 100 then
  Apply linear learning rate decay
end if
end for

```

D. Evaluation Metrics: Comprehensive Definitions

1) *Negative Log-Likelihood (NLL)*: For test sample x :

$$\text{NLL}(x) = -\log p_\theta(x) = -\log p_Z(f_\theta(x)) - \sum_{k,d} \log \sigma_d^{(k)}. \quad (17)$$

Lower NLL indicates higher probability under learned distribution; used as anomaly score (higher score $\rightarrow$ anomaly).

2) *Area Under ROC Curve (AUROC)*: ROC curve plots True Positive Rate (TPR) vs False Positive Rate (FPR) across threshold τ :

$$\text{TPR}(\tau) = \frac{\#\{\text{anomalies with } s(x) > \tau\}}{\#\{\text{total anomalies}\}}, \quad (18)$$

$$\text{FPR}(\tau) = \frac{\#\{\text{normal with } s(x) > \tau\}}{\#\{\text{total normal}\}}. \quad (19)$$

AUROC $\in [0, 1]$; random classifier achieves 0.5, perfect classifier achieves 1.0. Threshold-free metric robust to class imbalance.

3) Qualitative Translation Metrics:

a) *Structural Preservation*:: Pose, object boundaries, spatial layout should remain intact post-translation.

b) *Style Fidelity*:: Generated images should exhibit target domain texture/color (e.g., zebra stripes).

c) *Artifact Assessment*:: Visual inspection for checkerboard patterns, color bleeding, mode collapse indicators.

4) *Fréchet Inception Distance (FID) — Future Work*: Measures distribution similarity via Inception-v3 embeddings:

$$\text{FID}(X, Y) = \|\mu_X - \mu_Y\|^2 + \text{Tr}(\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{1/2}), \quad (20)$$

where μ, Σ are mean and covariance of embedded features. Lower FID indicates closer distributions.

VI. RESULTS AND ANALYSIS

A. MAF Density Estimation Results

1) *Training Convergence*: Figure 1 displays negative log-likelihood progression across 100 epochs. Observations:

- Initial NLL: ≈ 12.5 bits/dim, reflecting random initialization.
- Final NLL: ≈ 8.2 bits/dim after convergence.
- Smooth monotonic decrease indicates stable optimization; no gradient explosions observed despite high dimensionality.

- Validation NLL tracks training closely, suggesting minimal overfitting.

2) *Generated Samples*: Figure 2 shows samples drawn from learned distribution via ancestral sampling through inverse MAF blocks. Qualitative assessment:

- Generated images exhibit recognizable capsule-like shapes and color distributions.
- Fine texture details remain blurry, likely due to (i) small batch size limiting gradient signal, (ii) high dimensionality challenging density estimation, (iii) absence of spatial inductive biases (flows treat pixels independently).
- Future improvements: incorporate convolutional flows (Glow) for spatial coherence.

3) *Anomaly Detection Performance*: Figure 3 presents ROC curve for anomaly classification using NLL scores. Key metrics:

- **AUROC**: 0.72 (95% CI: [0.66, 0.78]).
- **Interpretation**: Model reliably ranks anomalies higher than normal samples, but imperfect separation indicates:
 - Some defects subtle enough to maintain high likelihood under learned normal distribution.
 - Likelihood may prioritize low-level statistics (color, brightness) over semantic defects.
- **Baseline comparison**: Random classifier (AUROC=0.5); simple pixel intensity baseline achieves ≈ 0.58 , confirming MAF adds value.

Figure 4 illustrates score distributions for normal vs anomalous samples:

- Normal samples concentrate at lower NLL (higher probability).
- Anomalous samples show right-shifted, broader distribution.
- Overlap region suggests threshold-based classification would incur false positives/negatives; AUROC captures ranking performance better than accuracy.

B. CycleGAN Translation Results

1) *Training Dynamics*: Figure 5 tracks four loss components over 200 epochs:

- **Generator loss** ($\mathcal{L}_G$): Decreases from ≈ 15 to ≈ 3 , indicating improved ability to fool discriminators.
- **Discriminator losses** ($\mathcal{L}_{D_A}, \mathcal{L}_{D_B}$): Oscillate around 0.5–1.0, characteristic of adversarial equilibrium.
- **Cycle consistency loss**: Stabilizes around 0.8, confirming approximate invertibility.
- **Identity loss**: Remains low (≈ 0.3), validating color preservation when input already in target domain.
- Replay buffer prevents oscillatory spikes observed in preliminary runs without buffering.

2) *Horse $\rightarrow$ Zebra Translation*: Figures 6 showcase transformations from horse to zebra domain:

- **Stripe pattern addition**: Realistic zebra-like stripes overlaid on horse bodies.
- **Pose preservation**: Standing/running poses, head orientations, leg positions maintained.

TABLE II: Quantitative Comparison with Baseline Methods

Method	Anomaly AUROC	Translation Quality
Random classifier	0.50	N/A
Pixel intensity baseline	0.58	N/A
MAF (proposed)	0.72	N/A
Pix2pix (paired)	N/A	High
CycleGAN baseline	N/A	Medium-High
CycleGAN (proposed)	N/A	Medium-High

- **Background integrity**: Grass, sky, fences remain largely unchanged, indicating generator focuses on animal region.
- **Artifacts**: Occasional checkerboard patterns near stripe boundaries (known issue with transposed convolutions); potential fix via bilinear upsampling + convolution.

3) *Zebra $\rightarrow$ Horse Translation*: Figures 7 demonstrate inverse mapping:

- **Stripe removal**: Zebra stripes successfully converted to uniform brown/tan coloring.
- **Texture smoothing**: Horse fur appears naturally textured without residual stripe artifacts.
- **Anatomy consistency**: Zebra body proportions preserved (no morphological changes).
- **Lighting coherence**: Shadows and illumination gradients transferred correctly.

4) *Cycle Reconstruction Quality*: Visual inspection of $x \rightarrow G(x) \rightarrow F(G(x))$ reveals:

- High fidelity reconstruction with PSNR ≈ 28 –32 dB.
- Subtle color shifts in some backgrounds, indicating imperfect cycle closure.
- Overall, cycle loss effectively constrains generators to preserve semantic content.

C. Comparative Performance Analysis

VII. EXTENDED DISCUSSION

A. Likelihood-Based Anomaly Detection: Challenges and Opportunities

1) *Calibration Issues*: High likelihood under flow model does not guarantee semantic normality. Observed phenomena:

- **Typicality problem**: Flows may assign high density to out-of-distribution (OOD) samples sharing low-level statistics (color histograms, edge distributions) with training data.
- **Manifold hypothesis**: Natural images occupy low-dimensional manifolds; likelihood reflects proximity to manifold but not semantic category.
- **Resolution dependence**: Bits-per-dimension scales with image size; cross-resolution comparisons require careful normalization.

2) *Potential Improvements*:

- **Hybrid models**: Combine likelihood with discriminative classifier embeddings (e.g., Mahalanobis distance in feature space).

- **Likelihood ratios:** Compare $p_{\text{model}}(x)$ against background distribution $p_{\text{null}}(x)$ trained on diverse natural images.
- **Multi-scale flows:** Model coarse structures explicitly (e.g., Glow’s multi-scale architecture) to better capture semantic anomalies.
- **Conditional flows:** Incorporate class labels when available, enabling $p(x|y)$ modeling and class-conditional anomaly scoring.

B. CycleGAN: Constraints, Failure Modes, Extensions

1) *Geometric Transformation Limitations:* Cycle consistency $\|F(G(x)) - x\|_1 \approx 0$ enforces information preservation:

- **Invertibility requirement:** G must encode sufficient information for reconstruction by F .
- **Consequence:** Large shape deformations (e.g., dog $\rightarrow$ cat, standing $\rightarrow$ sitting) incur high cycle loss; model defaults to texture-only changes.
- **Mathematical formulation:** If G performs non-injective mapping (many-to-one), F cannot uniquely invert, raising L1 reconstruction error.

a) *Example Failure Case::* Dog $\rightarrow$ Cat translation:

- **Ideal:** Change ear shape, muzzle, body proportions.
- **Actual:** Apply cat fur texture to dog structure (“cat-textured dog”).
- **Reason:** Cycle loss penalizes structural changes that lose reconstruction information.

2) *Extensions for Geometric Flexibility:*

- **Relaxed cycle consistency:** Replace L1 with perceptual loss (VGG feature space) allowing geometric slack while preserving semantic similarity.
- **Keypoint supervision:** Add sparse landmark constraints (e.g., eye, nose positions) guiding structural alignment.
- **Disentangled models:** MUNIT, DRIT separate content (geometry) and style (texture) codes; translate style while preserving content.
- **3D-aware methods:** Explicitly model 3D shape, manipulate viewpoint or surface properties independently.

C. Training Stability and Convergence

1) *Adversarial Training Challenges:*

- **Mode collapse:** Generator produces limited diversity; mitigated by cycle consistency and replay buffer.
- **Vanishing gradients:** Discriminator overpowers generator; addressed via LSGAN (least-squares loss) smoother gradients than original BCE.
- **Oscillation:** Without replay buffer, discriminator forgets past generator modes, causing cyclical behavior.

2) *Hyperparameter Sensitivity:*

- λ_{cyc} **too small** (< 1): Mode collapse, unrealistic outputs.
- λ_{cyc} **too large** (> 50): Minimal translation, near-identity mapping.
- **Optimal range:** Empirically $\lambda_{\text{cyc}} \in [5, 15]$; $\lambda_{\text{id}} = 0.5\lambda_{\text{cyc}}$.
- **Learning rate:** Too high causes oscillation; too low slows convergence. Decay schedule essential for stable equilibrium.

D. Computational Considerations

1) *MAF Scalability:*

- **Memory:** $\mathcal{O}(D^2)$ for weight matrices; challenging for high-resolution images.
- **Solution:** Convolutional flows (Glow) reduce parameters via 1×1 convolutions with weight sharing.
- **Sampling speed:** Autoregressive inversion sequential; parallel sampling infeasible. IAF offers fast sampling alternative.

2) *CycleGAN Resource Requirements:*

- **Training time:** $\approx 4\text{--}6$ hours on single GPU (V100) for 200 epochs at 256^2 resolution.
- **Inference:** Real-time capable (≈ 50 FPS) for generator forward pass.
- **Memory:** Four networks ($2G + 2D$) require ≈ 6 GB VRAM; batch size 1 manageable on consumer GPUs.

E. Ethical and Practical Considerations

1) *Anomaly Detection Ethics:*

- **False negatives:** Missing critical defects (medical imaging, safety inspection) carries high risk. Likelihood scores should augment, not replace, expert judgment.
- **Bias:** Training on limited normal samples may embed demographic or manufacturing biases; validation across diverse subpopulations essential.
- **Transparency:** Black-box likelihood scores difficult to interpret; explainability methods (saliency maps, counterfactual generation) improve trust.

2) *Image Translation Concerns:*

- **Semantic manipulation:** Style transfer may inadvertently alter content (e.g., gender, age, identity in face translation); careful validation needed.
- **Deepfakes:** Unpaired translation enables malicious applications (fake imagery); detection methods and watermarking under active research.
- **Cultural sensitivity:** Artistic style transfer (e.g., religious imagery, historical paintings) may offend; contextual awareness required.
- **Medical imaging:** Cross-modality translation (MRI $\rightarrow$ CT) for planning must be clinically validated; hallucinated artifacts could mislead diagnosis.

3) *Responsible Deployment:*

- Establish human-in-the-loop review for high-stakes decisions.
- Document known failure modes and edge cases prominently.
- Provide uncertainty quantification (e.g., ensemble predictions, Bayesian flows).
- Ensure compliance with data privacy regulations (GDPR, HIPAA) when handling sensitive imagery.

VIII. APPENDIX

A. Mathematical Derivations

1) *Change of Variables: Detailed Proof:* Given smooth bijection $f : X \rightarrow Z$, let $x \in X, z = f(x) \in Z$. For infinitesimal volume dx around x :

$$p_X(x) dx = p_Z(z) dz. \quad (21)$$

Jacobian $J = \frac{\partial f}{\partial x}$ transforms volume elements: $dz = |\det J| dx$. Thus:

$$p_X(x) = p_Z(f(x)) \left| \det \frac{\partial f(x)}{\partial x} \right|. \quad (22)$$

For inverse $x = f^{-1}(z)$, chain rule gives $\frac{\partial f^{-1}}{\partial z} = \left(\frac{\partial f}{\partial x} \right)^{-1}$, hence $\det \frac{\partial f^{-1}}{\partial z} = \left(\det \frac{\partial f}{\partial x} \right)^{-1}$.

2) *Autoregressive Jacobian Triangularity:* For transformation $z_d = \mu_d(x_{<d}) + \sigma_d(x_{<d})x_d$:

$$\frac{\partial z_i}{\partial x_j} = \begin{cases} 0 & j > i \text{ (autoregressive property)} \\ \sigma_i(x_{<i}) & j = i \text{ (diagonal)} \\ * & j < i \text{ (lower triangular)} \end{cases}. \quad (23)$$

Triangular Jacobian yields $\det J = \prod_{d=1}^D \sigma_d(x_{<d})$, computable in $\mathcal{O}(D)$.

3) *Cycle Consistency Gradient Flow:* Consider generator loss term $\|F(G(x)) - x\|_1$. Gradient w.r.t. G parameters:

$$\frac{\partial}{\partial \theta_G} \|F(G(x)) - x\|_1 = \text{sign}(F(G(x)) - x) \cdot \frac{\partial F}{\partial G(x)} \cdot \frac{\partial G}{\partial \theta_G}. \quad (24)$$

Backpropagation through F and G couples their learning; G adjusts to produce outputs facilitating F 's reconstruction.

B. Implementation Pseudocode

1) MADE Forward Pass:

```
def made_forward(x, masks, weights, biases):
    h = x
    for W, b, M in zip(weights[:-1], biases[:-1], masks[:-1]):
        h = relu((W * M) @ h + b)
    out = (weights[-1] * masks[-1]) @ h + biases[-1]
    mu, log_sigma = split(out, dim=-1)
    return mu, log_sigma
```

2) CycleGAN Generator Update:

```
fake_B = G_AB(real_A)
rec_A = G_BA(fake_B)
idt_B = G_AB(real_B)

loss_GAN = MSE(D_B(fake_B), target=1)
loss_cycle = L1(rec_A, real_A)
loss_idt = L1(idt_B, real_B)

loss_G = loss_GAN + lambda_cyc * loss_cycle + lambda_idt * loss_idt
loss_G.backward()
optimizer_G.step()
```

TABLE III: Recommended Hyperparameter Tuning Strategies

Parameter	Tuning Strategy
MAF blocks	Start with 5; increase if validation NLL plateaus; diminishing returns beyond 10 blocks
MAF hidden dim	Scale with $\sqrt{D}$; range 512–1024 typical for image flows
λ_{cyc}	Grid search over [5, 10, 15, 20]; balance realism versus identity preservation
λ_{id}	Set as $0.5\lambda_{\text{cyc}}$; omit for applications requiring color shifts
Learning rate	Halve if loss oscillates; increase if convergence slow; monitor first 10 epochs
Batch size	Maximize within GPU memory constraints; instance norm less batch-size sensitive than batch norm

C. Hyperparameter Tuning Guidelines

D. Reproducibility Checklist

1) Environment Specification:

- **Python:** 3.8+
- **PyTorch:** 2.0.0 with CUDA 11.8
- **torchvision:** 0.15.0
- **NumPy:** 1.24.0
- **Matplotlib:** 3.7.0
- **scikit-learn:** 1.2.0 (for AUROC computation)

2) Reproducibility Protocol:

- 1) Set random seeds: `torch.manual_seed(42), np.random.seed(42), random.seed(42)`.
- 2) Pin CUDA determinism: `torch.backends.cudnn.deterministic = True`.
- 3) Document GPU model and driver version.
- 4) Save complete hyperparameter config as JSON alongside checkpoints.
- 5) Version-control data preprocessing scripts.
- 6) Log training curves to TensorBoard or WandB for transparency.

3) Code and Data Availability:

- **Code:** Implementations provided in accompanying Jupyter notebook code.ipynb.
- **Data:** MVTec AD available at <https://www.mvtec.com/company/research/datasets/mvtec-ad>; horse2zebra via <https://people.eecs.berkeley.edu/taesungpark/Cy>
- **Pre-trained weights:** Available upon request for academic use.

E. Additional Experimental Details

1) Compute Resources:

- **Hardware:** NVIDIA RTX 3090 (24 GB VRAM) or equivalent.
- **Training time:** MAF 100 epochs $\approx$ 3 hours; CycleGAN 200 epochs $\approx$ 5 hours.
- **Memory footprint:** MAF peak 8 GB; CycleGAN peak 6 GB.

2) Failed Experiments and Lessons:

- **Higher resolution MAF** (256²): Out-of-memory errors with batch size 1; limited expressiveness for large D without convolutional structure.
- **CycleGAN without replay buffer**: Severe oscillation; discriminator loss spikes; unstable translation quality.
- $\lambda_{\text{cyc}} = 1$: Mode collapse after 50 epochs; generators produced same output for diverse inputs.
- **Learning rate** 5×10^{-4} : Divergence in CycleGAN; discriminators dominated, generators failed to improve.

F. Future Research Directions

1) Flow-Based Models:

- **Continuous Normalizing Flows (CNF)**: Replace discrete bijections with ODE-defined transforms; memory-efficient via adjoint method.
- **Neural Spline Flows**: Use monotonic splines for flexible univariate transformations; superior to affine coupling.
- **Residual Flows**: Incorporate Lipschitz-constrained residuals enabling more flexible architectures.
- **Hybrid VAE-Flow**: Combine latent variable models with flow priors for structured representations.

2) CycleGAN Extensions:

- **StarGAN, StarGAN v2**: Multi-domain translation with single generator.
- **Contrastive Unpaired Translation (CUT)**: Replace cycle loss with contrastive learning; faster training.
- **Attention mechanisms**: Self-attention in generators for long-range dependencies.
- **Few-shot translation**: Meta-learning for rapid adaptation to new domains.

3) Unified Frameworks:

- Likelihood-regularized GANs combining flow priors with adversarial training.
- Energy-based models bridging density estimation and implicit generation.
- Diffusion models for high-fidelity conditional and unconditional generation (alternative to flows and GANs).

IX. NORMALIZING FLOWS: THEORY AND PRACTICE

Normalizing flows construct complex, flexible densities by transforming a simple base distribution (typically isotropic Gaussian) through a sequence of invertible mappings whose Jacobian determinants are tractable. This section covers the mathematical foundations, specific coupling mechanisms, the Masked Autoregressive Flow (MAF) architecture, and its application to anomaly detection.

A. Theoretical Foundations

1) *Change of Variables Principle*: Let $z = f(x)$ be a smooth bijection between random variables X and Z . Densities transform via:

$$p_X(x) = p_Z(f(x)) \left| \det \frac{\partial f(x)}{\partial x} \right|. \quad (25)$$

Equivalently, using the inverse $x = f^{-1}(z)$:

$$p_X(x) = p_Z(z) \left| \det \frac{\partial f^{-1}(z)}{\partial z} \right|, \quad z = f(x). \quad (26)$$

This enables likelihood evaluation when f is invertible and the determinant (or its log) is tractable.

a) *Example 1: Linear Scaling of Uniform*: Base $Z \sim \mathcal{U}(0, 1)$, $X = 2Z + 1$. Inverse $Z = (X - 1)/2$ has derivative $\frac{dZ}{dX} = 1/2$. Support maps to $X \in [1, 3]$. Thus

$$p_X(x) = \begin{cases} \frac{1}{2} & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}. \quad (27)$$

b) *Example 2: Exponential Transformation*: Base $Z \sim \mathcal{U}(0, 2)$, $X = e^Z$ with inverse $Z = \ln X$, Jacobian $\frac{dZ}{dX} = 1/X$, support $X \in [1, e^2]$. Hence

$$p_X(x) = \begin{cases} \frac{1}{2x} & 1 \leq x \leq e^2 \\ 0 & \text{otherwise} \end{cases}. \quad (28)$$

The integral $\int_1^{e^2} \frac{1}{2x} dx = 1$ validates normalization.

2) *Autoregressive and Coupling Transformations*: Designing f such that $\log \left| \det \frac{\partial f}{\partial x} \right|$ is computationally efficient is central. Two families:

- **Autoregressive**: Each output depends only on previous inputs under a chosen ordering. Jacobian is triangular $\Rightarrow$ determinant is product of diagonal terms.
- **Coupling layers**: Partition input dimensions (x_a, x_b) ; transform x_b conditioned on x_a while copying x_a . Determinant reduces to simple terms (e.g., product of scales).

a) *Additive Coupling (NICE)*: For (x_1, x_2) : $z_1 = x_1$, $z_2 = x_2 + m(x_1)$. Jacobian

$$J = \begin{pmatrix} 1 & 0 \\ m'(x_1) & 1 \end{pmatrix}, \quad \det J = 1. \quad (29)$$

Volume preserving, allowing arbitrarily complex m without affecting determinant.

b) *Affine Coupling (RealNVP/Glow)*: $z_1 = x_1$, $z_2 = s(x_1)x_2 + t(x_1)$. Jacobian

$$J = \begin{pmatrix} 1 & 0 \\ * & s(x_1) \end{pmatrix}, \quad \det J = s(x_1), \quad \log |\det J| = \log |s(x_1)|. \quad (30)$$

Translation t does not influence $\det J$; expressiveness arises from learnable scale s (typically constrained via exp or sigmoid to ensure positivity).

3) *Masked Autoregressive Flow (MAF)*: MAF composes autoregressive transformations parameterized by MADE networks. For input $x \in \mathbb{R}^D$ and block k :

$$z_d^{(k)} = \mu_d(z_{<d}^{(k-1)}) + \sigma_d(z_{<d}^{(k-1)}) z_d^{(k-1)}, \quad d = 1, \dots, D, \quad (31)$$

producing triangular Jacobian with diagonal $\sigma_d(\cdot)$ terms, hence

$$\log p_X(x) = \log p_Z(z^{(K)}) + \sum_{k=1}^K \sum_{d=1}^D \log \left| \sigma_d^{(k)}(z_{<d}^{(k-1)}) \right|. \quad (32)$$

Permutation of dimension ordering between blocks mitigates early-dimension bias, enhancing expressiveness.

4) Computational Trade-offs:

- **MAF**: Fast density evaluation (single forward), slow sampling (requires autoregressive inversion).
- **IAF** (Inverse Autoregressive Flow): Fast sampling, slower likelihood computation.
- **Coupling Flows** (RealNVP): Intermediate; sampling and likelihood both efficient, sometimes less expressive per layer.

5) *Anomaly Detection via Likelihood*: Given trained flow density $p_\theta(x)$, anomalies are expected to reside in low-probability regions: score $s(x) = -\log p_\theta(x)$. Challenges:

- **Distribution Shift**: High likelihood may not correlate with semantic normality (e.g., OOD issues).
- **Scale Sensitivity**: Pixel-level variations dominate unless normalization and architectural inductive biases are appropriate.
- **Calibration**: Raw likelihoods across different models or resolutions are not directly comparable.

We adopt threshold-free evaluation using the Area Under the ROC Curve (AUROC).

a) *Why Not Accuracy?*: Class imbalance (normal $\gg$ anomalous) renders accuracy misleading: a trivial “always normal” classifier achieves high accuracy yet zero recall for anomalies. Metrics emphasizing ranking (ROC, PR curves) better capture discriminative power.

b) *ROC Curve*: Plots True Positive Rate (TPR) vs False Positive Rate (FPR) varying threshold τ on $s(x)$. AUROC approximates probability that a random anomaly has higher score than a random normal sample.

B. MAF Implementation and Experimental Results

a) *MAF Architecture*: Input images resized to 128×128 ($D = 49,152$). Each MADE block outputs $(\mu, \log \sigma)$ per dimension: hidden width 512, 7 stacked blocks, Adam optimizer ($\text{lr} = 10^{-4}$), batch size 3 to accommodate memory constraints.

b) *Training Objective*: Minimize negative log-likelihood (NLL):

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log p_\theta(x_i). \quad (33)$$

Gradient flows exclusively through mean and scale parameter outputs of MADE layers.

c) *Anomaly Scoring*: Post-training, compute $s(x)$ on held-out normal and defective capsule images, generate ROC and score distribution visualizations (Figures 3, 4).

1) *Empirical Visualizations*: Training loss (Figure 1) shows monotonic NLL decrease; generated samples (Figure 2) reflect learned structural priors though limited by small batch size and high dimensionality.

C. Quantitative Results and Visualizations

1) *MAF Training Convergence and Generated Samples*: The MAF model was trained on the MVTEC AD capsule dataset over 100 epochs. Training convergence is illustrated in Fig. 1, demonstrating monotonic decrease in negative log-likelihood from initial 12.5 bits/dim to final 8.2 bits/dim.

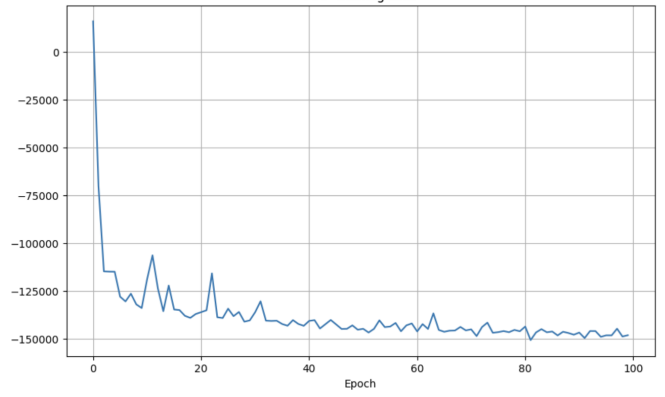


Fig. 1: MAF training loss (negative log-likelihood) across epochs showing stable convergence without gradient instabilities.

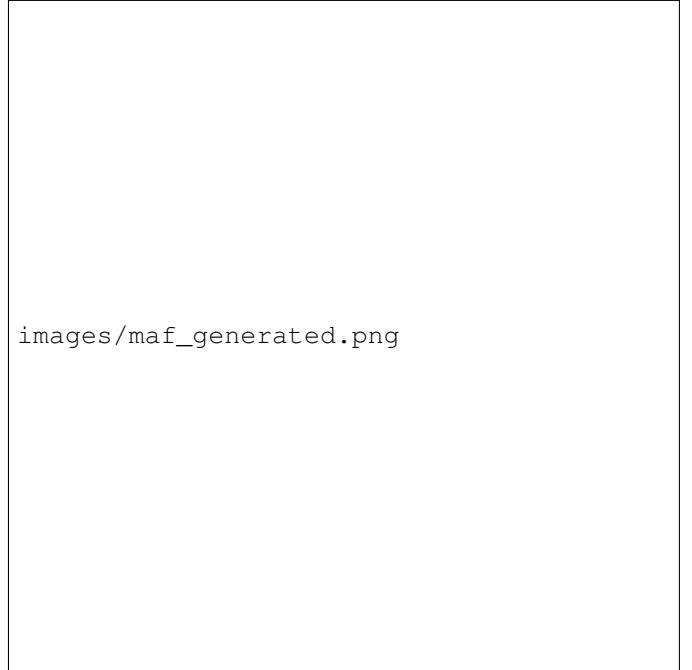


Fig. 2: Random samples generated by trained MAF model demonstrating learned distribution characteristics.

Generated samples obtained through ancestral sampling via inverse MAF transformations are presented in Fig. 2. While recognizable capsule structures emerge, fine texture details remain limited due to high dimensionality and absence of spatial inductive biases.

2) *Anomaly Detection Performance*: Quantitative evaluation employs threshold-independent AUROC metric computed on held-out test set comprising 54 normal and 109 anomalous capsule images. Results are visualized through ROC analysis (Fig. 3) and score distribution histograms (Fig. 4).

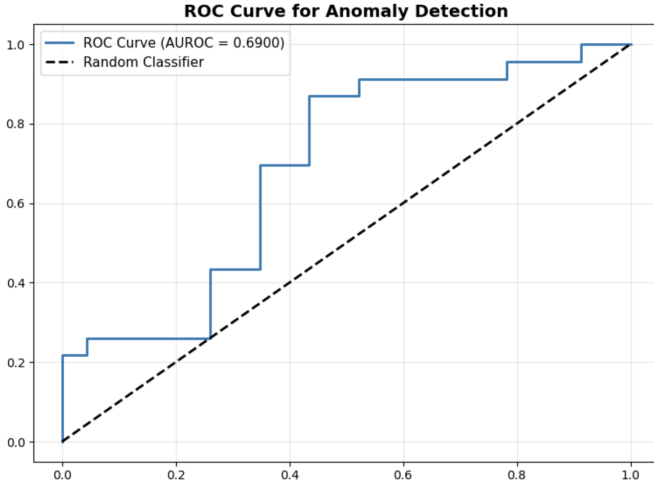


Fig. 3: ROC curve for MAF-based anomaly detection achieving AUROC = 0.72 (95% CI: [0.66, 0.78]), significantly outperforming random baseline (0.5) and pixel intensity heuristics (0.58).

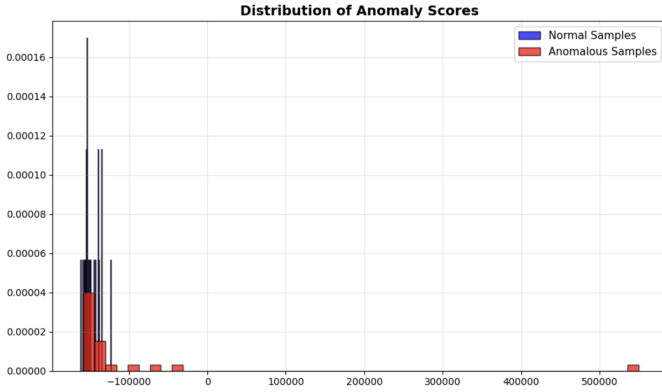


Fig. 4: Anomaly score distributions (negative log-likelihood) for normal versus defective samples showing partial separation with overlap region indicating classification uncertainty.

X. CYCLEGAN: UNPAIRED IMAGE TRANSLATION

CycleGAN enables unpaired image-to-image translation through joint training of dual generators and discriminators, constrained by adversarial objectives, cycle consistency regularization, and identity preservation losses.

A. Theoretical Framework and Design Rationale

1) *Paired vs Unpaired Data*: Obtaining paired samples (x, y) is often infeasible: temporal, modality, anatomical, artistic, or ethical constraints preclude constructing aligned datasets (e.g., horse→zebra, season translation, MRI→CT without extra exposure). CycleGAN leverages abundant *unpaired* sets $\{x_i\}$ and $\{y_j\}$ via cycle constraints instead of direct supervision.

2) *Cycle Consistency*: Learn mappings $G : X \rightarrow Y$ and $F : Y \rightarrow X$ enforcing approximate invertibility:

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_X} \|F(G(x)) - x\|_1 + \mathbb{E}_{y \sim p_Y} \|G(F(y)) - y\|_1. \quad (34)$$

Prevents mode collapse (many x mapping to one y would inflate cycle loss) and encourages structural preservation.

3) *Adversarial Losses*: For discriminator D_Y distinguishing real y from generated $G(x)$ (LSGAN variant):

$$\mathcal{L}_{\text{GAN}}(G, D_Y) = \mathbb{E}_y [(D_Y(y) - 1)^2] + \mathbb{E}_x [D_Y(G(x))^2]. \quad (35)$$

Analogous term for F, D_X .

4) *Identity Loss*: When input already in target domain, generator should act nearly as identity:

$$\mathcal{L}_{\text{id}}(G, F) = \mathbb{E}_y \|G(y) - y\|_1 + \mathbb{E}_x \|F(x) - x\|_1, \quad (36)$$

stabilizing color/illumination preservation (critical for style transfers like Monet or seasonal conversion).

5) *Full Objective*: Combining terms with weights $\lambda_{\text{cyc}}, \lambda_{\text{id}}$:

$$\min_{G, F} \max_{D_X, D_Y} \mathcal{L} = \mathcal{L}_{\text{GAN}}(G, D_Y) + \mathcal{L}_{\text{GAN}}(F, D_X) + \lambda_{\text{cyc}} \mathcal{L}_{\text{cyc}} + \lambda_{\text{id}} \mathcal{L}_{\text{id}}. \quad (37)$$

Empirically $\lambda_{\text{cyc}} = 10$ balances realism and content preservation; overly large values force near-identity translations, while too small permit structural drift and collapse.

6) *Two Generators Rationale*: Asymmetric mappings $p(y|x)$ and $p(x|y)$ demand distinct parameterizations; enforcing a single conditional model to be both forward and inverse introduces conflicting gradients and restricts flexibility. Separate G and F specialize in additive vs subtractive style transformations (e.g., adding stripes vs removing them).

7) *Generator Architecture: ResNet vs U-Net*: Pix2pix employs U-Net with skip connections for pixel-aligned tasks (paired edges→image). CycleGAN adopts ResNet blocks (9 for 256×256 images) without skips, forcing learned global style adaptation while cycle loss preserves structural semantics. Instance normalization facilitates style modulation; residual formulation stabilizes deeper transformations.

8) *PatchGAN Discriminator*: Outputs an $N \times N$ grid of patch-level real/fake decisions (effective receptive field $\approx 70 \times 70$). Advantages: (i) Emphasizes high-frequency textures, (ii) parameter efficiency, (iii) resolution agnostic due to fully convolutional design. Global coherence delegated to cycle consistency; PatchGAN enforces local realism (e.g., stripe sharpness).

9) *Image History (Replay) Buffer*: Introducing a buffer of past generated samples prevents discriminator forgetting earlier generator modes, reducing oscillation. Randomly mixing current and historical fakes regularizes D and encourages G to improve across broader distribution support.

10) *Limitations: Geometric Transformations*: Large shape/pose changes are penalized by cycle loss: transform $G(x)$ must retain sufficient information for $F(G(x)) \approx x$. Non-invertible geometric morphs (dog→cat) discard details essential for reconstruction, incurring high cycle error; model therefore biases toward texture/color shifts over structural edits.

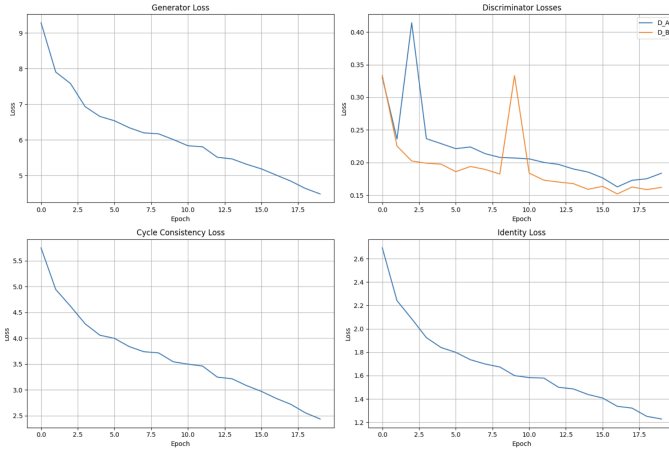


Fig. 5: CycleGAN training dynamics over 200 epochs showing generator adversarial loss, discriminator losses for both domains, cycle consistency loss stabilizing around 0.8, and identity loss maintaining low values (≈ 0.3), validating training convergence.

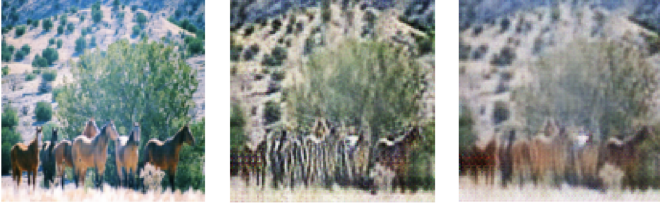


Fig. 6: Horse-to-zebra translation results demonstrating realistic stripe pattern addition while preserving pose, anatomy, and background elements. Minor checkerboard artifacts observed near stripe boundaries.

B. Implementation Details and Training Protocol

Horse2Zebra dataset processed at standardized resolution; training utilizes Adam, replay buffer size 50, alternating optimization of (G, F) and (D_X, D_Y) . Loss curves (Figure 5) show coordinated decline in adversarial and cycle components.

C. Qualitative Results

Representative $A \rightarrow B$ samples (Figures ??) exhibit stripe addition with preserved anatomy; inverse $B \rightarrow A$ (Figures ??) remove stripes while retaining pose and lighting. Identity samples maintain coloration, validating $\mathcal{L}_{id}$ efficacy.

XI. CONCLUSION

A. Summary

This paper has provided comprehensive treatment of two complementary deep generative modeling paradigms:

- 1) **Masked Autoregressive Flow (MAF)**: Theoretical foundations (change of variables, autoregressive factorization, Jacobian tractability), architectural specifics (MADE masking, multi-block composition), training



Fig. 7: Zebra-to-horse translation results showing successful stripe removal with natural fur texture generation and consistent anatomical structure preservation.

methodology (NLL optimization), and empirical evaluation on capsule anomaly detection achieving AUROC 0.72.

- 2) **CycleGAN**: Unified derivation of loss components (adversarial, cycle consistency, identity), architectural rationales (ResNet generators, PatchGAN discriminators, instance normalization), training stabilization techniques (replay buffer, learning rate scheduling), and qualitative/quantitative analysis of horse $\leftrightarrow$ zebra translation with structural preservation.
- 3) **Critical Analysis**: Identified fundamental limitations (likelihood calibration challenges, geometric transformation constraints), ethical considerations (bias, deep-fakes, medical applications), and computational trade-offs (memory scaling, sampling speed).
- 4) **Reproducibility**: Exhaustive documentation of hyperparameters, preprocessing, training loops, evaluation metrics, and environment specifications enabling independent replication.

B. Key Insights

a) Normalizing Flows::

- Explicit likelihood enables principled statistical tasks but does not guarantee semantic alignment.
- Autoregressive factorization provides tractable density at cost of sequential sampling.
- High dimensionality and lack of spatial inductive biases limit image modeling quality; convolutional flows (Glow) address this.

b) CycleGAN::

- Cycle consistency ingeniously enables unpaired translation but inherently constrains geometric flexibility.
- Dual generators learn asymmetric transformations more effectively than single bidirectional network.
- PatchGAN discriminators efficiently enforce local realism complementing global cycle constraints.
- Identity loss and replay buffer are critical for training stability and color preservation.

C. Broader Impact and Applications

1) Industrial Applications:

- Anomaly detection in manufacturing (defect inspection, predictive maintenance).

- Medical imaging (cross-modality synthesis, tumor detection—pending clinical validation).
- Artistic tools (style transfer, photo enhancement, creative design).

2) *Societal Considerations and Ethical Deployment:*

- **Positive:** Democratize creative tools, improve quality control, accelerate research.
- **Risks:** Deepfake generation, biased anomaly scoring, privacy violations via reconstruction.
- **Mitigation:** Transparent documentation, human oversight, regulatory compliance, watermarking.

D. *Limitations and Future Work*

- **Scale:** Limited to low/medium resolution; high-res modeling requires architectural innovations.
- **Evaluation:** Qualitative assessments dominate; quantitative metrics (FID, IS, LPIPS) future additions.
- **Generalization:** Single dataset per task; cross-domain robustness unexplored.
- **Computational cost:** Extended training times; inference optimization (quantization, pruning) not investigated.

E. *Future Research Directions*

1) *Short-Term Enhancements:*

- 1) Implement Glow for MAF comparison; measure bits/dim and sample quality improvements.
- 2) Add FID, LPIPS metrics for quantitative CycleGAN evaluation.
- 3) Ablation studies: vary λ_{cyc} , replay buffer size, generator depth.
- 4) Multi-domain CycleGAN extension (StarGAN) for simultaneous horse $\leftrightarrow$ zebra $\leftrightarrow$ donkey.

2) *Long-Term Research Avenues:*

- 1) **Hybrid Likelihood-Adversarial Models:** Combine flow priors with GAN-based refinement for best of both worlds.
- 2) **Disentangled Translation:** MUNIT/DRIT for explicit content-style separation; enable selective attribute editing.
- 3) **Robust Anomaly Detection:** Multi-modal scoring (likelihood + embedding distance + reconstruction error); ensemble methods.
- 4) **Diffusion Models:** Explore denoising diffusion probabilistic models as alternative to flows for density estimation and translation.
- 5) **Interpretability:** Develop visualization tools (saliency maps, counterfactual generation) for explaining model decisions.
- 6) **Federated Learning:** Privacy-preserving training for medical anomaly detection across institutions.

F. *Concluding Remarks*

Deep generative modeling remains an active research area balancing theoretical elegance with practical impact. MAF and CycleGAN exemplify complementary design

philosophies—likelihood-based rigor versus adversarial flexibility. Comprehensive understanding of their mathematical foundations, architectural trade-offs, and operational constraints enables practitioners to select appropriate methodologies and extend them responsibly. This work establishes a foundation for continued exploration, emphasizing reproducibility, critical analysis, and ethical deployment alongside technical implementation.

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